# CHAPTER I

# INTRODUCTION TO STUDY

## 1.1 INTRODUCTION

A College is a specialized institution within the educational system where students receive higher education and training in a wide range of academic and professional fields. Colleges serve as essential platforms for intellectual growth, career development, and personal transformation. They offer structured programs that lead to undergraduate and postgraduate degrees, diplomas, and certificates across various disciplines such as science, arts, commerce, engineering, business, medicine, and more. The structure and focus of colleges can differ significantly; some function as general institutions offering a broad curriculum, while others are more specialized, focusing on areas such as technology, liberal arts, or vocational training. In addition to academic learning, colleges play a central role in developing critical life skills, promoting innovation, and encouraging independent thinking and creativity.

Colleges are not only centres of education but also of research, innovation, and community engagement. Many institutions collaborate with industries, government bodies, and other academic organizations to conduct research, solve real-world problems, and contribute to societal progress. College campuses are vibrant communities that support extracurricular involvement through clubs, sports, cultural events, and student leadership opportunities. These activities help students build confidence, social skills, and a strong sense of identity and purpose. Furthermore, colleges play a significant role in shaping future leaders, entrepreneurs, scientists, educators, and professionals who go on to make impactful contributions across society.

In today's increasingly digital and data-driven educational environment, data analysis has become a vital component in the management and advancement of colleges. Educational institutions generate vast amounts of data daily, including student enrolment records, academic performance, attendance, faculty evaluations, resource usage, financial transactions, and campus services. By applying advanced data analytics, colleges can extract meaningful insights that support decision-making, enhance student learning outcomes, and improve institutional efficiency.

Data analysis helps colleges in a variety of ways. For example, it can be used to track student performance trends, identify at-risk students, and personalize learning experiences to meet diverse academic needs. Predictive analytics can forecast enrolment numbers, allowing colleges to better manage resources such as faculty assignments, classroom space, and academic materials. Institutions also use analytics to evaluate course effectiveness, monitor faculty productivity, and assess the impact of different teaching methodologies.

Operationally, data can inform strategic planning by analysing financial data, optimizing budget allocation, and ensuring efficient use of infrastructure. Analytics also plays a crucial role in improving student services such as counselling, placement, and library access by understanding usage patterns and student satisfaction. Colleges can also monitor key performance indicators (KPIs) like graduation rates, dropout rates, job placement rates, and student engagement metrics.

## 1.2 OBJECTIVES

- To provide quality higher education across various academic and professional disciplines.
- To promote critical thinking, creativity, and innovation among students.
- To prepare students for successful careers by developing both theoretical knowledge and practical skills.
- To contribute to research and development through academic inquiry and industry collaboration.
- To foster holistic development through extracurricular, cultural, and leadership activities.
- To in still values of ethics, responsibility, and citizenship in students.

## 1.3 SCOPE

- **Academic Development** – Offering diverse programs across disciplines like science, arts, commerce, technology, and more.
- **Research and Innovation** – Encouraging faculty and students to engage in research, publish papers, and innovate.
- **Skill Enhancement** – Providing skill-based training, workshops, and certifications to improve employability.
- **Personal Growth** – Fostering soft skills, leadership qualities, and self-confidence through campus activities.
- **Career Preparation** – Offering placement support, internships, and industry interactions.
- **Cultural and Social Exposure** – Promoting cultural diversity, community outreach, and social responsibility.

## 1.4 IMPORTANCE

- This helps colleges make informed, data-driven decisions to improve education quality.
- Assists in strategic academic planning, resource management, and policy formulation.
- Detects academic or operational issues early for timely interventions.
- Contributes to improved student performance and success outcomes.
- Enhances understanding of student behaviour and learning patterns.
- It improves institutional efficiency by analysing resource usage and operations.

## 1.5 RESEARCH METHODOLOGY

1. **Data Collection** – The dataset was obtained from college records including student demographics, academic performance, attendance, faculty details, and course information.
2. **Data Cleaning and Pre-processing** – Missing entries, duplicate records, and formatting inconsistencies were resolved to maintain data accuracy and reliability.
3. **Data Analysis** – Descriptive statistics and trend analysis were conducted to identify patterns in student performance, dropout rates, and faculty effectiveness.
4. **Data Visualization** – Graphs, charts, and dashboards were created to visually present insights related to academic trends, resource utilization, and student behaviour.
5. **Interpretation of Results** – Analytical outcomes and visual data representations were interpreted to derive meaningful conclusions for improving academic planning and college operations.

# CHAPTER II

# SYSTEM REQUIREMENTS

## 2.1 HARDWARE REQUIREMENTS

**COMPUTER**

- **Monitor**

  It displays information about the current work on screen.

- **Mouse**

  It allows users to control the movement of a pointer on the screen, making it easier to interact with graphical elements and perform various actions.

- **Keyboard**

  It enables you to input text, execute commands, navigate, and perform various tasks through key presses

- **Printer**

  It prints the record.

- **RAM SIZE**- 6GB

- **ROM SIZE**-512GB

## 2.2 SOFTWARE REQUIREMENTS

- **Operating System**- Windows 11
- **Data Visualization Tool**-Python (Dash, Plotly, Matplotlib, Seaborn)
- **Programming language**- Python
- **Development Environment-** Jupyter Notebook
- **Data Processing Libraries**- Pandas, NumPy
- **Excel/CSV Support**- Microsoft CSV

## 2.3 TECHNOLOGIES USED

### 1. Programming Language

Python: The primary language used for data manipulation, analysis, and visualization due to its simplicity and rich library ecosystem.

### 2. Development Environment

Jupyter Notebook: An interactive web-based environment ideal for writing and executing Python code, visualizing data, and documenting the analysis process.

### 3. Key Python Libraries

Pandas: For data loading, cleaning, and manipulation (e.g., filtering rows, grouping data, handling missing values).

NumPy: For numerical computations and array operations.

Matplotlib: For basic data visualizations like bar charts, line plots, and pie charts.

Seaborn: For advanced statistical visualizations with enhanced aesthetics.

Plotly : For interactive visualizations.

### 4. Data Source

Excel files exported from a CSV File system or downloaded from public repositories like Kaggle.

### 5. Data Visualization

Graphs and charts were created using Matplotlib and Seaborn to explore trends, identify sales peaks, and evaluate the effectiveness of marketing campaigns.

# CHAPTER III

# INTRODUCTION TO ORGANIZATION

## 3.1 INTRODUCTION OF ORGANIZATION

Shivraj College, Gadhinglaj was established in 2005 in Gadhinglaj, Maharashtra. Over the years, it has become one of the most prominent educational institutions in the region. The college is committed to providing high-quality education and fostering the academic and personal growth of its students. With a dedicated team of experienced faculty and modern infrastructure, Shivraj College offers a wide range of programs across various disciplines, including Arts, Commerce, and Science.

The college aims to provide a student-centered learning environment, promoting not only academic excellence but also the overall development of its students. It places great emphasis on encouraging students to participate in extracurricular activities, which play a vital role in their holistic development. Shivraj College continuously strives to maintain a high standard of education and campus facilities, ensuring that students have access to the best resources for learning.

One of the key strengths of Shivraj College is its focus on providing personalized attention to students. The institution believes in creating an inclusive learning environment, where each student is supported in their academic journey. The college also offers modern facilities such as a well-equipped library, digital classrooms, sports facilities, and a variety of student support services, ensuring that students have all the tools they need to succeed.

**3.2 SEVICES OFFERED**

- Undergraduate and postgraduate courses in Arts, Commerce, and Science.
- Skill-building workshops and certification programs for enhanced employability.
- Career scounselling and placement assistance for student career development.
- Extracurricular activities including sports, cultural events, and leadership programs.
- Personalized academic support, mentoring, and guidance for students.
- Encouragement of research and innovation to foster academic growth.

**3.3 MILESTONE ACHIVEVED**

- Established a state-of-the-art infrastructure with modern classrooms and laboratories.
- Introduced postgraduate programs in various disciplines to expand academic offerings.
- Trained faculty in innovative teaching methodologies and digital tools to enhance learning.
- Expanded the library facilities with a wide range of books, journals, and online resources.
- Upgraded sports facilities, encouraging more students to participate in athletic activities.
- Upgraded sports facilities, encouraging more students to participate in athletic activities.

## 3.4 MISSION AND VISION

### VISION:

"Our vision is to create a world class higher education system that empowers our students to become leaders and change-makers in their communities and beyond. By providing a rigorous and transformative education that fosters critical thinking, ethical reasoning and a deep sense of social responsibility, we aim to produce a new generation of leaders who are committed to building a strong and just India. Together, we will build a strong and prosperous India based on reasoning, conscience, compassion and philanthropy."

### MISSION:

"Our mission is to empower the students of Gadhinglaj, Ajara, and Chandgad tahsil in Maharashtra, Karnataka and Goa-Kokan border areas to achieve their full potential and contribute to the development of their communities. We believe that quality education is the key to transforming minds and improving socioeconomic outcomes and we are committed to providing our students with access to the latest knowledge and technologies. By fostering a culture of excellence, innovation and social responsibility, we aim to create a brighter future for all."

## 3.5 FUTURE PLANS OF THE ORGANIZATION

- Expand infrastructure with new buildings, classrooms, and modern facilities.
- Introduce new undergraduate and postgraduate programs in emerging fields.
- Promote research and innovation through dedicated research centers and collaborations.
- Strengthen community engagement through social responsibility initiatives and outreach programs.
- Invest in advanced digital tools for online learning and virtual classrooms.
- Foster stronger industry collaborations for internships, placements, and career development.
- Organize workshops, seminars, and cultural activities to enhance student experiences.

# CHAPTER IV

# DATA ANALYSIS AND INTERPRETATION

**Input**

```
import pandas as pd
file_path = r"C:\Users\HP\Documents\College Data.xlsx"
df = pd.read_excel(file_path)
df.head()
```

**Output**

**SHIVRAJ COLLEGE, GADHINGLAJ**

| | Name of the Student | Gender | Course Pursued | Fees | Location | Degree |
|---|---|---|---|---|---|---|
| 0 | Akolkar Prachi Subhash | Female | BBA | 14500 | Gadhinglaj | Bachelor's |
| 1 | Bilawar Bhumika Anand | Female | BBA | 14500 | Uttur | Bachelor's |
| 2 | Biranje Yogita Vishnu | Female | BBA | 14500 | Nesari | Bachelor's |
| 3 | Chougale Arpita Kashinath | Female | BBA | 14500 | Chandgad | Bachelor's |
| 4 | Chougale Arun Suresh | Male | BBA | 14500 | Ajra | Bachelor's |

**Input**

```
print(df.columns)
```

**Output**

```
Index(['Name of the Student ', 'Gender ', 'Course Pursued', 'Fees ',
       'Location ', 'Degree'],
      dtype='object')
```

**Input**

```
import pandas as pd
file_path = r"C:\Users\HP\Documents\College Data.xlsx"
df = pd.read_excel(file_path)
df.columns = df.columns.str.strip()
print(df.columns.tolist())
print(df['Gender'].value_counts())
```

**Output**

```
['Name of the Student', 'Gender', 'Course Pursued', 'Fees', 'Location', 'Degree']
Gender
Male                    897
Female                  862
Female                  202
Female                    4
Name: count, dtype: int64
```

**Input**

df.groupby('Course Pursued')['Fees'].mean()

**Output**

```
Course Pursued
B.COM                    8000.0
BBA                     14500.0
BCA                     24000.0
BCS                     25400.0
BSC ANIMATION          100000.0
BSC. FOOD SCIENCE       25000.0
M.A (ECONOMICS)         11000.0
M.A (ENGLISH)           11000.0
M.A (HINDI)             11000.0
M.A (MARATHI)           11000.0
M.COM                   10000.0
M.Sc. COMPUTER          45000.0
M.Sc.MICROBIOLOGY       25000.0
MA. (SOCIOLOGY)         11000.0
Name: Fees, dtype: float64
```

**Input**

df['Location'].value_counts()

**Output**

```
Location
Gadhinglaj    375
Uttur         343
Nesari        320
Chandgad      315
Ajra          304
Mahagaon      304
Chandgad        2
Nesari          2
Name: count, dtype: int64
```

**Input**

```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go

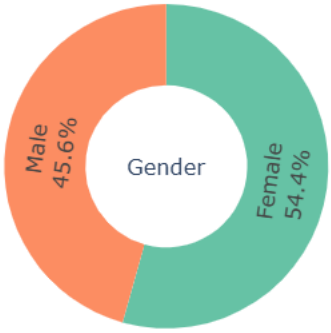
df = pd.read_excel(r"C:\Users\HP\Documents\College Data.xlsx")
df.columns = df.columns.str.strip()
df['Gender'] = df['Gender'].str.strip().str.title()
df['Location'] = df['Location'].str.strip()
df['Course Pursued'] = df['Course Pursued'].str.strip().str.upper()
df['Degree'] = df['Degree'].str.strip().str.upper()

gender_counts = df['Gender'].value_counts().reset_index()
gender_counts.columns = ['Gender', 'Count']
fig1 = px.pie(
    gender_counts, values='Count', names='Gender',

   hole=0.5, title='Gender Distribution',

  color_discrete_sequence=px.colors.qualitative.Set2
)
fig1.update_traces(textinfo='percent+label')
fig1.update_layout(annotations=[dict(text='Gender', x=0.5, y=0.5, showarrow=False)])
fig1.show()
```

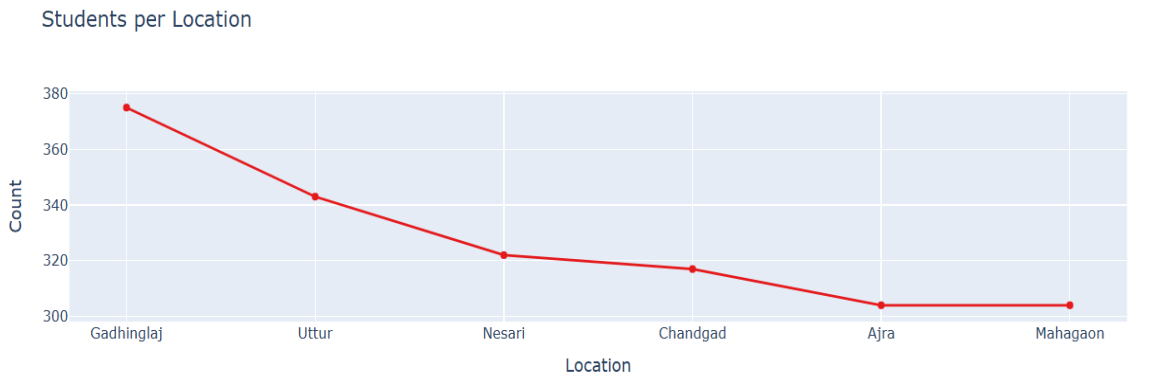
**Output**




**Input**

```
location_counts = df['Location'].value_counts().reset_index()
location_counts.columns = ['Location', 'Count']
fig2 = px.line(
    location_counts, x='Location', y='Count',
    markers=True, title='Students per Location',
    color_discrete_sequence=px.colors.qualitative.Set1
)
fig2.show()
```

**Output**



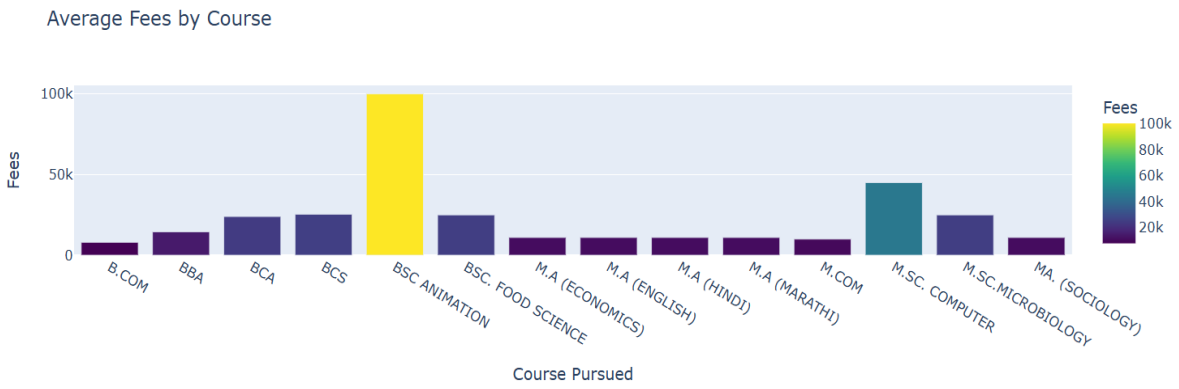
**Input**

```
fees_by_course = df.groupby('Course Pursued')['Fees'].mean().reset_index()

fig3 = px.bar(
    fees_by_course,
    x='Course Pursued',

   y='Fees',
    title='Average Fees by Course',
    color='Fees',
    color_continuous_scale='Viridis'
)
fig3.show()
```

**Output**

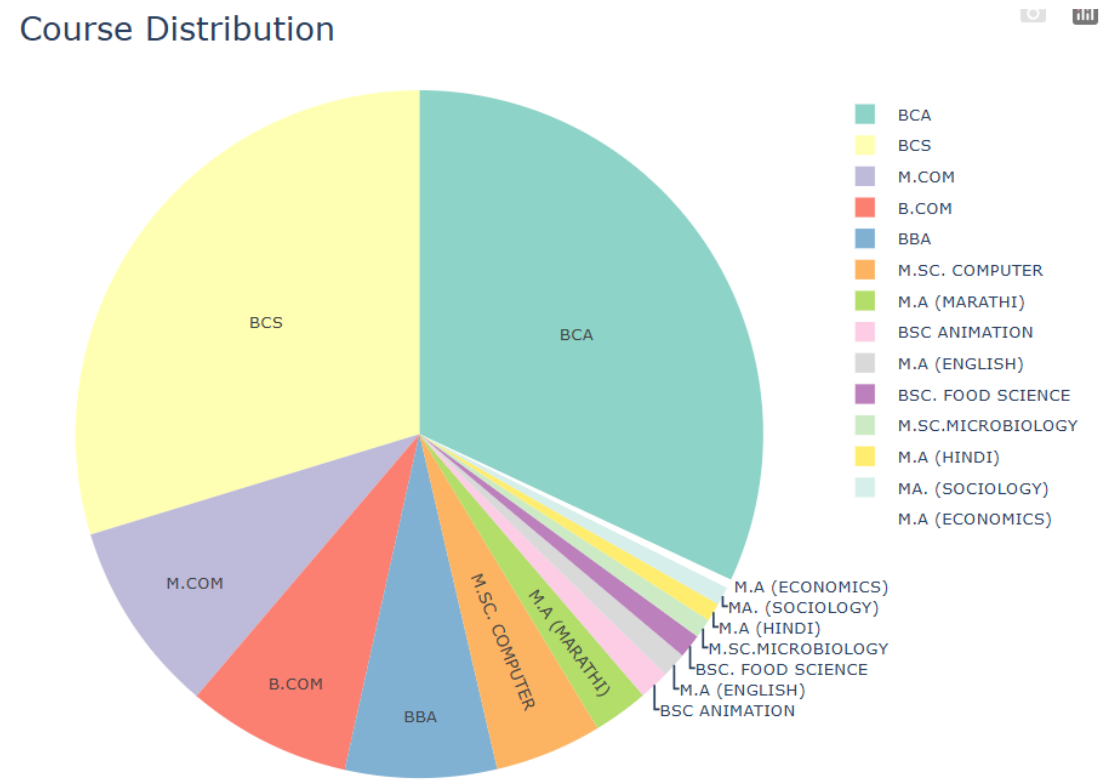


**Input**

```
course_counts = df['Course Pursued'].value_counts().reset_index()
course_counts.columns = ['Course', 'Count']
fig4 = px.pie(
    course_counts,
    values='Count',
    names='Course',
    title='Course Distribution',
    color_discrete_sequence=px.colors.qualitative.Set3
)
fig4.update_layout(
    title_font_size=20,
    legend_font_size=10,
    margin=dict(t=60, b=20, l=20, r=20),
    height=500,
    width=700
)
fig4.update_traces(
    textinfo='label',
    textfont_size=10
)

fig4.show()
```

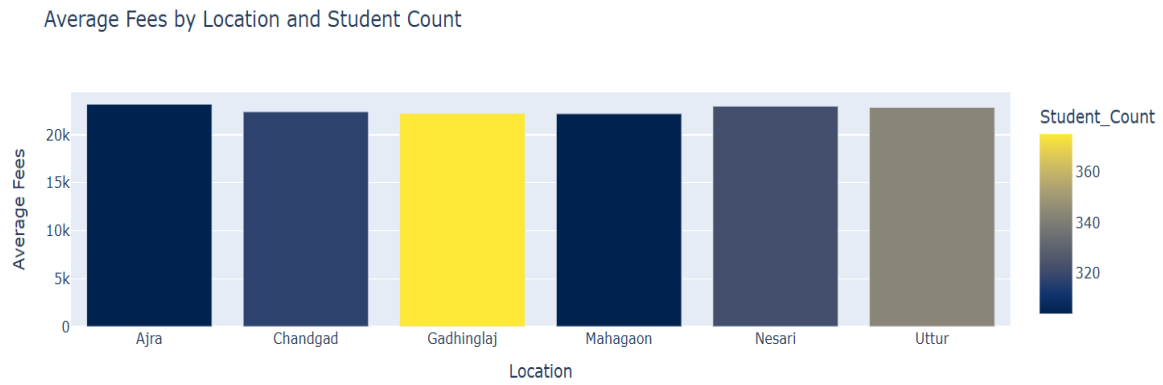
**Output**




**Input**

```
agg_location = df.groupby('Location').agg(
    Avg_Fees=('Fees', 'mean'),
    Student_Count=('Location', 'count')
).reset_index()
fig5 = px.bar(
    agg_location,
    x='Location',
    y='Avg_Fees',
    color='Student_Count',
    title='Average Fees by Location and Student Count',
    labels={'Avg_Fees': 'Average Fees'},
    color_continuous_scale='Cividis'
)
fig5.show()
```

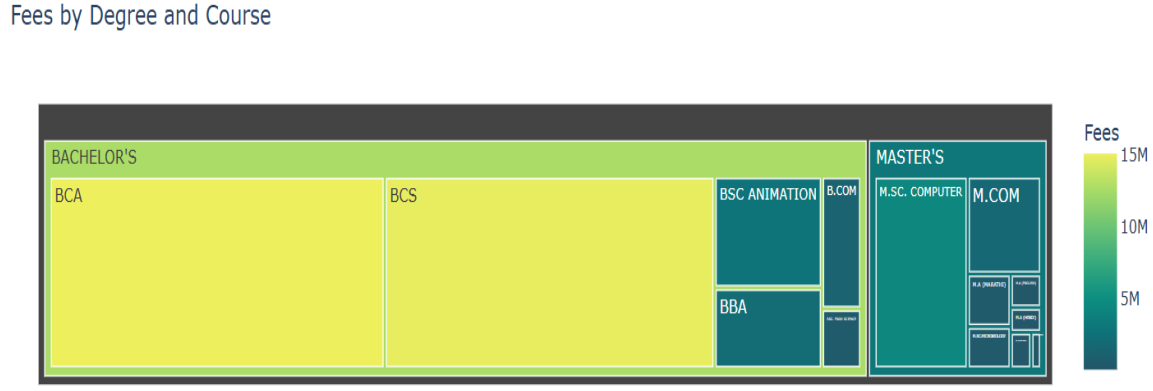
**Output**




**Input**

```
treemap_data = df.groupby(['Degree', 'Course Pursued'])['Fees'].sum().reset_index()
fig6 = px.treemap(
    treemap_data,
    path=['Degree', 'Course Pursued'],
    values='Fees',
    title='Fees by Degree and Course',
    color='Fees',
    color_continuous_scale='Aggrnyl'
)
fig6.show()
```

**Output**



**Input of Dashboard**

```
import dash
from dash import dcc, html
import plotly.express as px
import pandas as pd

app = dash.Dash(__name__)
server = app.server

gender_data = pd.DataFrame({
    'Gender': ['Male', 'Female'],
    'Count': [456, 544]
})

location_data = pd.DataFrame({
    'Location': ['Gadhinglaj', 'Uttur', 'Nesari', 'Chandgad', 'Ajra', 'Mahagaon'],
    'Count': [375, 343, 322, 316, 305, 305]
})

fees_data = pd.DataFrame({
    'Course': ['B.COM', 'BBA', 'BCA', 'BCS', 'BSC ANIMATION', 'BSC FOOD SCIENCE',
               'M.A (ECONOMICS)', 'M.A (ENGLISH)', 'M.A (HINDI)', 'M.A (MARATHI)',
'M.COM', 'M.SC COMPUTER', 'M.SC.MICROBIOLOGY', 'M.A (SOCIOLOGY)'],
    'Fees': [10000, 15000, 25000, 26000, 100000, 12000, 11000, 11000, 11000, 11000, 20000,
45000, 25000, 11000]
})

course_dist_data = pd.DataFrame({
    'Course': ['BCA', 'BCS', 'M.COM', 'B.COM', 'BBA', 'M.SC COMPUTER',
```

```
 'M.A (MARATHI)', 'BSC ANIMATION', 'M.A (ENGLISH)',
            'BSC FOOD SCIENCE', 'M.SC.MICROBIOLOGY', 'M.A (HINDI)', 'M.A (SOCIOLOGY)', 'M.A (ECONOMICS)'],
    'Count': [300, 290, 100, 90, 80, 60, 30, 20, 10, 10, 10, 10, 10, 10]
})

location_fees_data = pd.DataFrame({
    'Location': ['Ajra', 'Chandgad', 'Gadhinglaj', 'Mahagaon', 'Nesari', 'Uttur'],
    'Avg_Fees': [23000, 22000, 22000, 22000, 22500, 22700],
    'Student_Count': [305, 316, 375, 305, 322, 343]
})

degree_course_fees_data = pd.DataFrame({
    'Degree': [
        "BACHELOR'S", "BACHELOR'S", "BACHELOR'S", "BACHELOR'S", "BACHELOR'S",
        "MASTER'S", "MASTER'S", "MASTER'S", "MASTER'S", "MASTER'S",
        "MASTER'S", "MASTER'S", "MASTER'S", "MASTER'S"
    ],
    'Course': [
        'BCA', 'BCS', 'BSC ANIMATION', 'B.COM', 'BBA',
        'M.SC COMPUTER', 'M.COM', 'M.A (MARATHI)', 'M.A (ENGLISH)', 'M.A (HINDI)',
        'M.A (SOCIOLOGY)', 'M.A (ECONOMICS)', 'M.SC.MICROBIOLOGY', 'BSC FOOD SCIENCE'
    ],
    'Fees': [
        15000000, 14000000, 5000000, 2000000, 3000000,
        8000000, 6000000, 1500000, 800000, 800000,
        700000, 700000, 900000, 1000000
    ]
})
```

```
total_students = gender_data['Count'].sum()

total_courses = course_dist_data['Course'].nunique()

male_count = gender_data.loc[gender_data['Gender'] == 'Male', 'Count'].values[0]

female_count = gender_data.loc[gender_data['Gender'] == 'Female', 'Count'].values[0]

gender_ratio = round(male_count / female_count, 2)

average_fees = fees_data['Fees'].mean()


gender_fig = px.pie(gender_data, names='Gender', values='Count', hole=0.5)

gender_fig.update_traces(textinfo='percent+label', marker=dict(colors=['#fc8d62',
'#66c2a5']))

gender_fig.update_layout(title_text='Gender Distribution', title_x=0.5)


location_fig = px.line(location_data, x='Location', y='Count', markers=True)

location_fig.update_traces(line_color='red')

location_fig.update_layout(title='Students per Location', xaxis_title='Location',
yaxis_title='Count')


fees_fig = px.bar(fees_data, x='Course', y='Fees', color='Fees',
color_continuous_scale='Viridis')

fees_fig.update_layout(title='Average Fees by Course', xaxis_title='Course Pursued',
yaxis_title='Fees')


course_dist_fig = px.pie(course_dist_data, names='Course', values='Count')

course_dist_fig.update_layout(title='Course Distribution')


location_fees_fig = px.bar(location_fees_data, x='Location', y='Avg_Fees',
color='Student_Count', color_continuous_scale='Cividis')

location_fees_fig.update_layout(title='Average Fees by Location and Student Count',
yaxis_title='Average Fees')
```

```
treemap_fig = px.treemap(degree_course_fees_data, path=['Degree', 'Course'], values='Fees', color='Fees', color_continuous_scale='Viridis')

treemap_fig.update_layout(title='Fees by Degree and Course')

app.layout = html.Div([

   html.H1('Student Dashboard', style={'textAlign': 'center', 'color': '#003366', 'marginBottom': '20px'}),


      html.Div([

      html.Div([

         html.H3('Total Students', style={'color': '#555'}),

         html.H2(f"{total_students}", style={'color': '#003366'})

      ], className='card'),


      html.Div([

         html.H3('Total Courses', style={'color': '#555'}),

         html.H2(f"{total_courses}", style={'color': '#003366'})

      ], className='card'),


      html.Div([

         html.H3('Male : Female Ratio', style={'color': '#555'}),

         html.H2(f"{gender_ratio}", style={'color': '#003366'})

      ], className='card'),


      html.Div([

         html.H3('Average Fees (₹)', style={'color': '#555'}),

         html.H2(f"{average_fees:.2f}", style={'color': '#003366'})

      ], className='card'),

   ], style={'display': 'flex', 'justifyContent': 'space-around', 'marginBottom': '40px'}),
```

```
        html.Div([

    dcc.Graph(figure=gender_fig, style={'width': '48%', 'display': 'inline-block'}),
        dcc.Graph(figure=location_fig, style={'width': '48%', 'display': 'inline-block'}),
    ]),

    html.Div([
        dcc.Graph(figure=fees_fig, style={'width': '100%'}),
    ]),

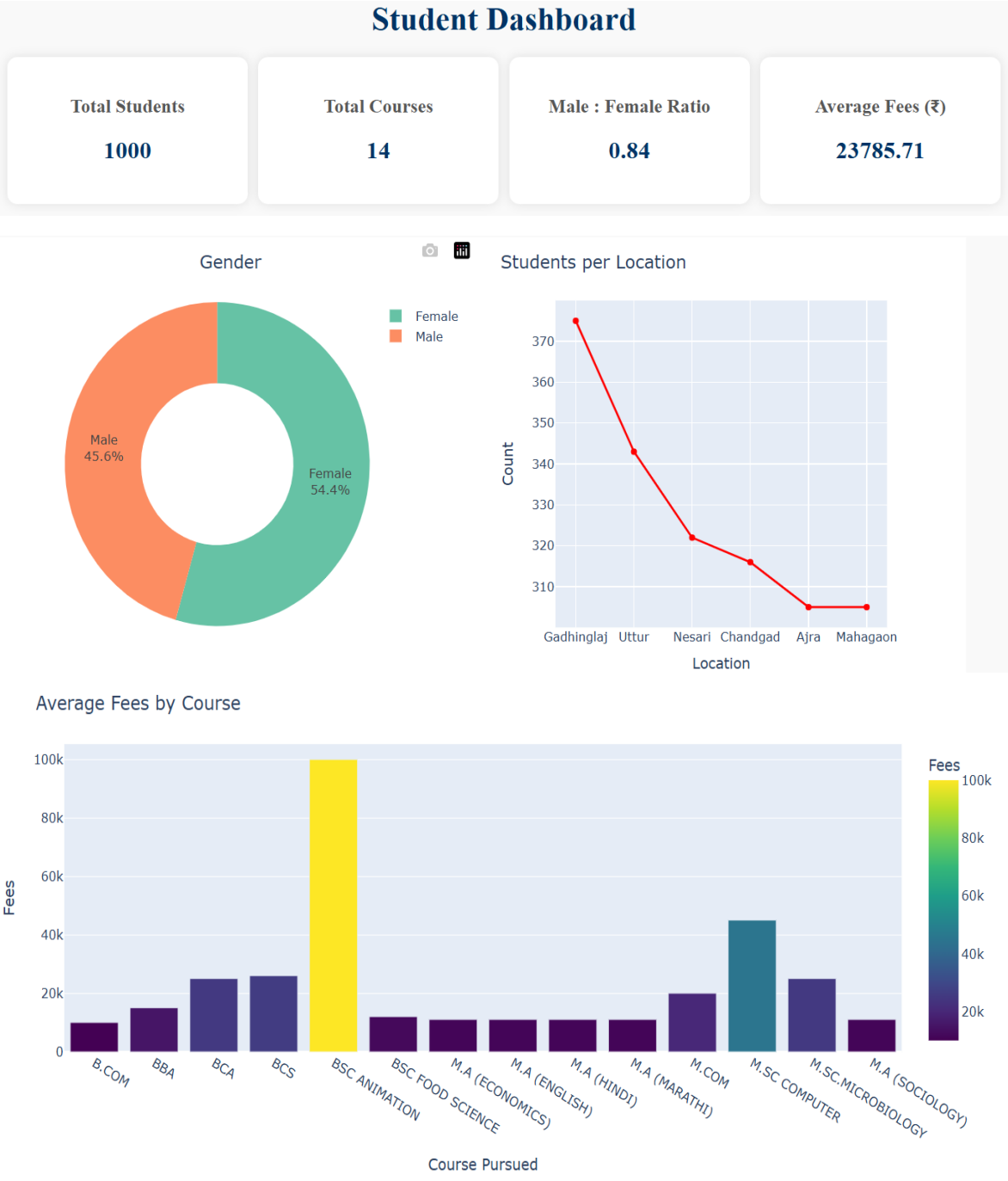
    html.Div([
        dcc.Graph(figure=course_dist_fig, style={'width': '48%', 'display': 'inline-block'}),
        dcc.Graph(figure=location_fees_fig, style={'width': '48%', 'display': 'inline-block'}),
    ]),

    html.Div([
        dcc.Graph(figure=treemap_fig, style={'width': '100%'}),
    ]),
], style={'backgroundColor': '#f9f9f9', 'padding': '20px'})

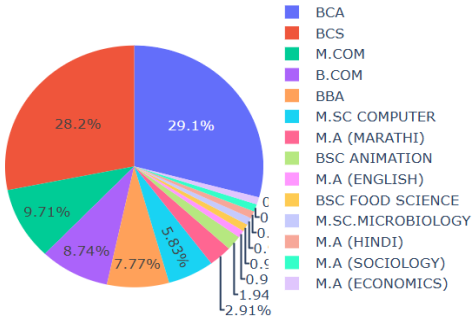
app.index_string = '''
<!DOCTYPE html>
<html>
    <head>
        {%metas%}
        <title>Student Dashboard</title>
        {%favicon%}
        {%css%}
        <style>
            .card {
```

```
            background: white;
            padding: 20px;
            box-shadow: 0px 0px 10px rgba(0,0,0,0.1);
            border-radius: 10px;
            width: 20%;
            text-align: center;
        }
    </style>
</head>
<body>
    {%app_entry%}
    <footer>
        {%config%}
        {%scripts%}
        {%renderer%}
    </footer>
</body>
</html>
'''
if __name__ == '__main__':
    app.run(debug=True, port=8051)
```

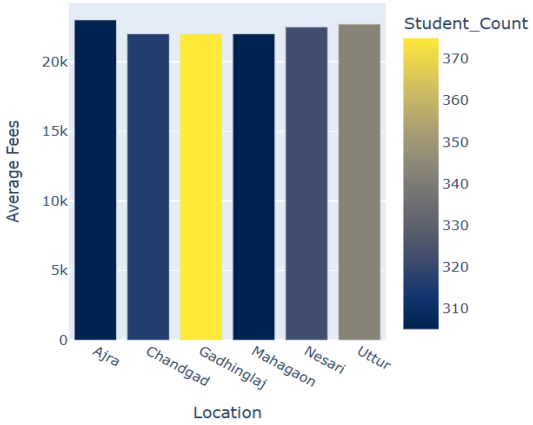
## Output of Dashboard



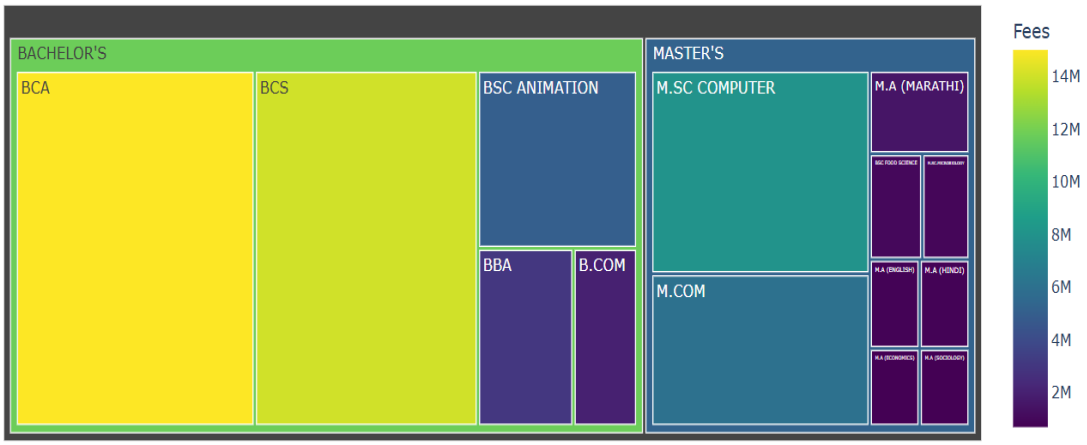
Course Distribution




Average Fees by Location and Student Count

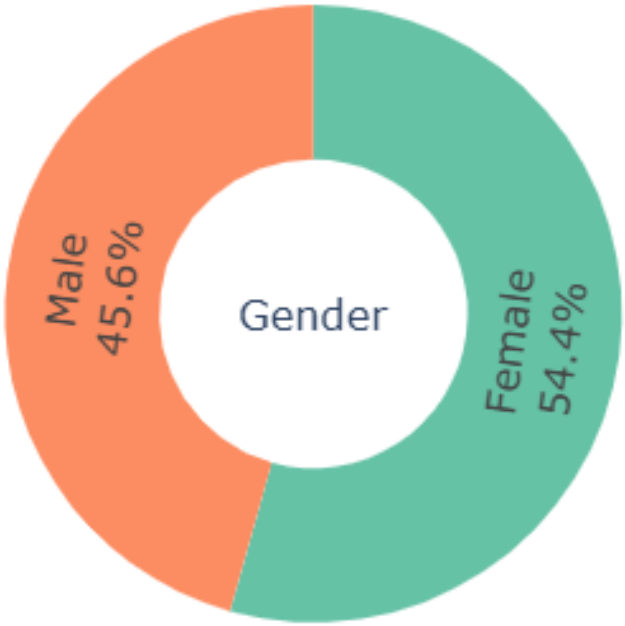



Fees by Degree and Course



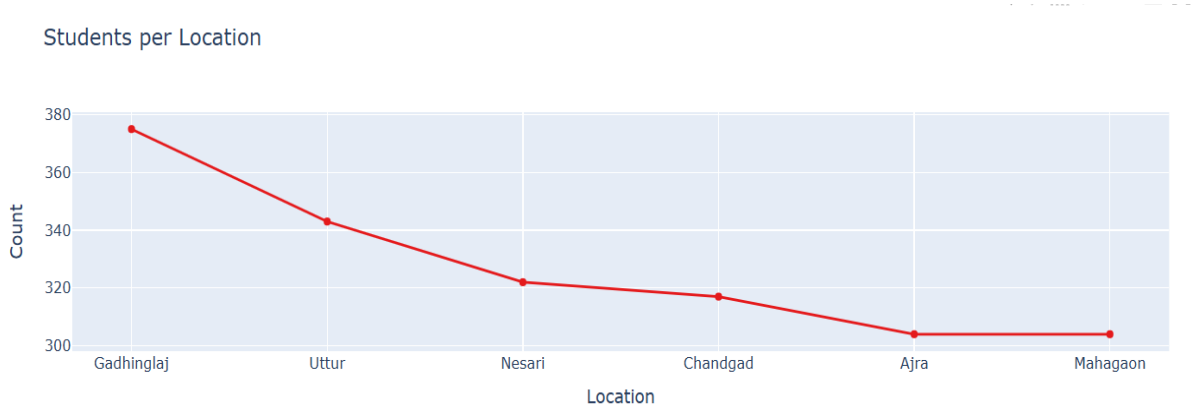
## DATA INTERPRETATION

### 4.1 GRAPH



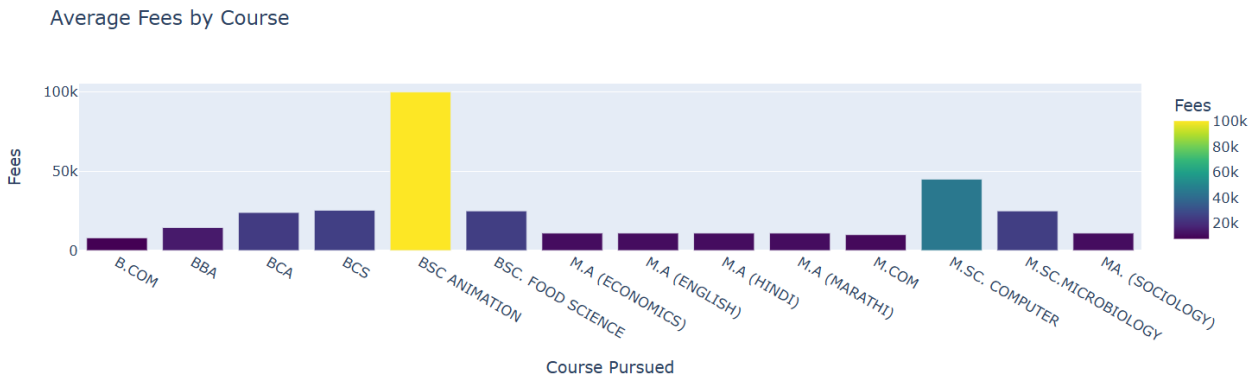

The gender distribution reveals that 54.4% of the students are female, while 45.6% are male. This indicates a slightly higher enrollment of female students compared to males. The difference, though not very large, highlights growing female participation. Overall, the gender ratio appears fairly balanced in the student population.

## 4.2 GRAPH



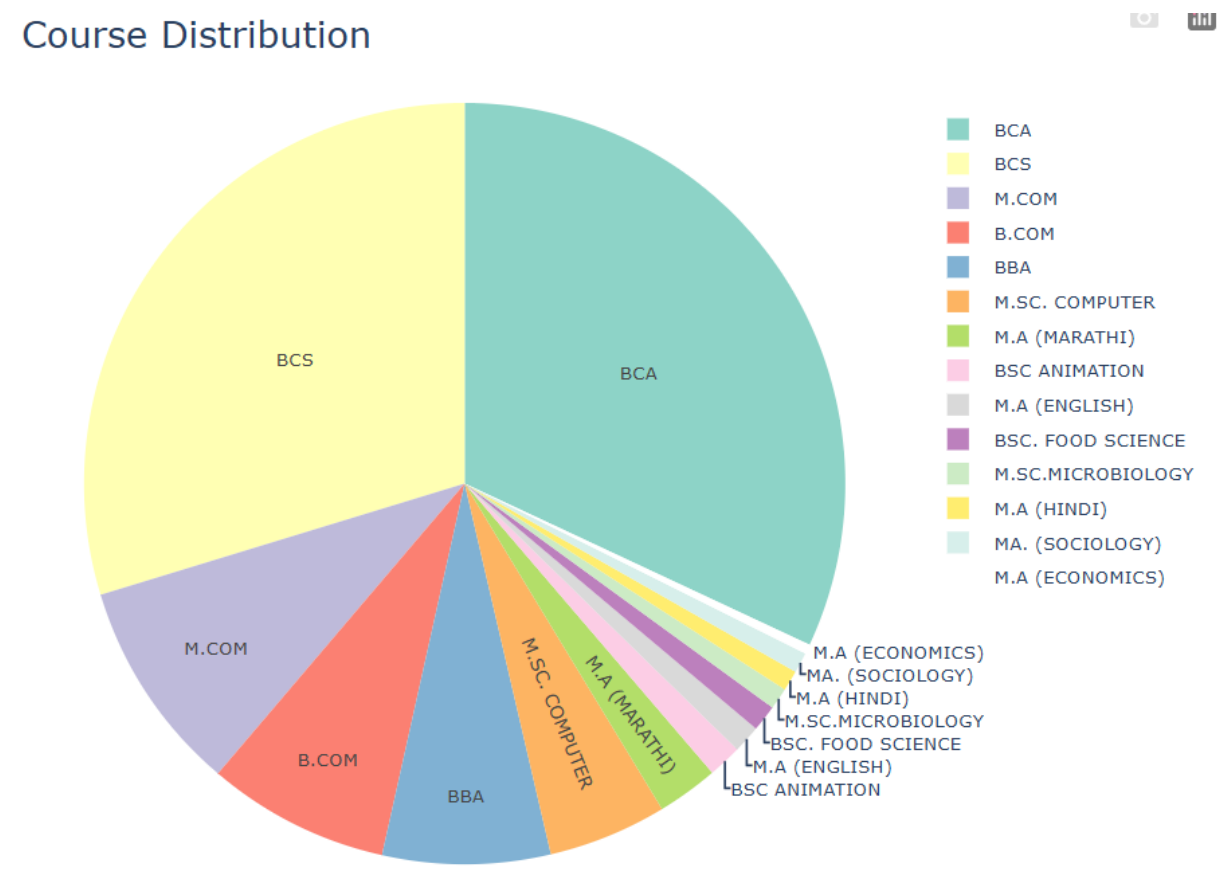

Gadhinglaj has the highest number of students, followed by Uttur and Nesari. There is a gradual decline in student numbers from Gadhinglaj to Ajra and Mahagaon. Ajra and Mahagaon show the lowest student counts among the listed locations. This trend highlights a concentration of students in more populated or accessible areas.

## 4.3 GRAPH



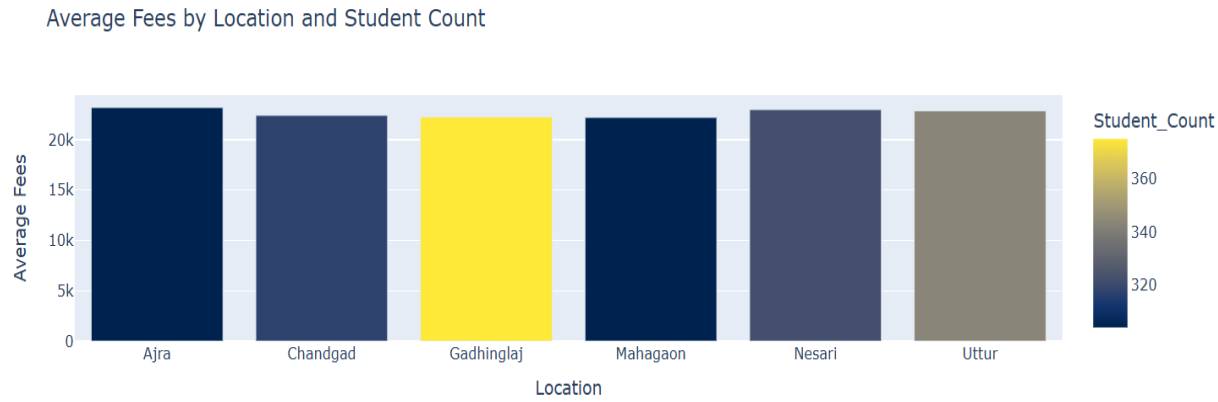

The B.Sc. Animation course has the highest average fees among all courses. M.Sc. Computer also has relatively higher fees compared to other programs. Courses like B.Com, BBA, and various M.A. programs have significantly lower fees. There is a wide fee variation, indicating differing resource demands among courses.

## 4.4 GRAPH



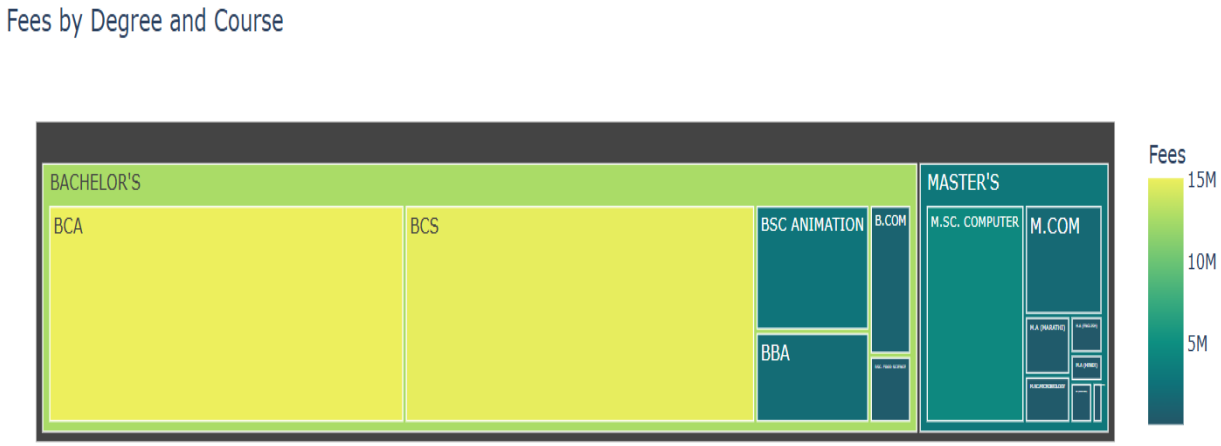

BCA and BCS courses clearly dominate the overall course distribution and they together form more than half of the total student base. This suggests a very strong student interest in computer applications and computer science fields.

## 4.5 GRAPH




Ajra and Chandgad have the highest average fees, while Gadhinglaj has the highest student count among the locations, suggesting that Gadhinglaj attracts more students despite slightly lower average fees.

## 4.6 GRAPH




Bachelor's degrees (especially BCA and BCS) dominate fee contributions, with BCA and BCS having the highest total fees, while Master's degrees like M.Sc. Computer and M.Com contribute moderately.

# CHAPTER V

# FINDINGS AND SUGGESTIONS

## FINDINGS

- BCA is the most popular course with the highest number of students.
- BCS is the second most popular course among students.
- Courses like M.Sc. Computer and M.Com also have good enrollment but much lower than BCA/BCS.
- BBA has moderate enrollment compared to BCA and BCS.
- Few courses like B.Sc. Animation and B.Sc. Food Science have very low student numbers.
- Average fees vary from location to location; some locations have higher average fees.
- Some courses have more balanced gender ratios, while others are male or female-dominated.

## SUGGESTIONS

- Promote less popular courses like B.Sc. Animation and Food Science through seminars and advertising.

- Focus marketing efforts in locations where fewer students are enrolling.

- Offer scholarships or fee discounts for high-fee locations to attract more students.

- Introduce new specializations under popular programs like BCA and BCS to keep students interested.

- Maintain a balance of male and female students by creating women-friendly initiatives in male-dominated courses.

- Improve placement services especially for courses with low enrollment to boost confidence.

- Expand collaborations with industries and companies for internships especially in technical fields.

# CHAPTER VI

# CONCLUSION

From the analysis of the college data, it is clear that technical courses like BCA and BCS are the most popular among students. Other courses like M.Sc. Computer and M.Com also attract a good number of students, but there are some courses that need more promotion as they have very few enrollments. Most students come from a Gadhinglaj locations, and the average fee also changes depending on the location.

To improve further, the college can focus on marketing less popular courses, offering fee discounts, and making career opportunities visible to students. Encouraging more female students in technical fields and building strong industry connections will also help in the long run. Overall, the college is doing well in technical education, and with a few improvements, it can grow even stronger and attract more students from different backgrounds.

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